

Roman Merkulov

Python Backend & AI Engineer

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Technical skills

Languages: Python, TypeScript/JavaScript, SQL, Java; working knowledge of Rust, Bash, and PowerShell

Backend and APIs: FastAPI, Flask, REST, OpenAPI, Pydantic, asynchronous processing, service integrations

Cloud and infrastructure: AWS S3, SQS, CloudWatch, DocumentDB; Docker, Linux, Git

Databases: PostgreSQL, MongoDB, Amazon DocumentDB, Redis, MariaDB

AI and data: PyTorch, scikit-learn, NumPy, Pandas, spaCy, transformer models, LLM APIs, prompt design, document classification, structured extraction

Frontend: React, TypeScript, JavaScript, HTML/CSS

Security: Burp Suite, web/API security testing, OAuth and JWT, OWASP vulnerability classes, threat modeling, recon automation

Professional experience

Software Engineer

Apr. 2026–Present

FSM Rechtsanwälte

Vienna, Austria

- Develop AWS-based components connecting Python backend services to React/TypeScript frontends.
- Evaluate LLM-based retrieval and document-review workflows to speed up initial document screening.
- Conduct basic technical and security reviews of websites as part of legal workflow screening.
- Improve database queries and application workflows that support document processing and screening.

Lead Software Engineer

Mar. 2025–Mar. 2026

Schweinhammer Rechtsanwalt

Vienna, Austria

- Designed and delivered an AI-assisted document-processing application with Python backend services, React/TypeScript interfaces, MongoDB/DocumentDB, and AWS.
- Built asynchronous OCR, document-classification, and structured-extraction workflows using Amazon SQS and commercial and open-source LLMs.
- Developed APIs, validation rules, state-management workflows, and interfaces for legal and procurement documents.
- Integrated external verification services and handled successful, expired, missing, and failed verification states explicitly.
- Owned technical architecture and implementation across backend, frontend, AI, and infrastructure components.

Freelance Software Engineer

May 2023–Jan. 2025

Fintech client project

Remote

- Developed Python web services and REST APIs for fintech and banking-related workflows.
- Built AWS-hosted application components and integrations between frontend applications, backend services, databases, and external providers.
- Integrated LLM-based functionality for document processing, user assistance, and application automation.
- Improved error handling, testing, and maintainability across cloud-hosted services.
- Supported deployment, service configuration, and production debugging of distributed application components.

Earlier experience

Machine Learning Intern NLABTEAM	Jan. 2022–Nov. 2022 Remote
- Developed and improved NLP pipelines for chatbot applications, including text preprocessing, intent classification, entity extraction, and response workflows. - Used spaCy and transformer-based models to improve natural-language-understanding components. - Evaluated chatbot output and diagnosed issues in classification, retrieval, and response relevance. - Tested model configurations and preprocessing strategies, then integrated ML components into the existing chatbot architecture.	
Technical Consultant Independent	Apr. 2020–Jun. 2022 Vienna / Remote
- Developed and optimized Python ETL pipelines for client data workflows. - Supported migration from monolithic applications to Docker- and Kubernetes-based service architectures. - Advised on backend architecture, infrastructure security, deployment practices, and secure data handling.	

Selected technical projects and research

Distributed Data-Processing Platform Rust, Python	Jul. 2026–Present
- Developing a backend service for uploading and querying CSV and telemetry datasets through an HTTP API. - Using Rust, Axum, Tokio, and Serde for asynchronous request handling, data validation, and structured JSON responses. - Using Python to generate test datasets, validate API results, and build integration and performance tests. - Extending the architecture with analytical query execution, background job processing, persistent metadata storage, and observable failure handling.	
Autonomous AI-agent security research	
- Research autonomous-agent security boundaries, including prompt injection, persistent-memory poisoning, unsafe tool invocation, and authorization failures. - Built an isolated ARM-based VM and Docker lab for vulnerability reproduction and defensive monitoring. - Applied STRIDE threat modeling to agent components, tools, memory, communication channels, and trust boundaries. - Reproduced published vulnerabilities and documented allow/deny policies, telemetry requirements, and security controls.	

Education

B.Sc. Computer Science University of Vienna Bachelor thesis: Security of Autonomous AI Agents: prompt injection, memory poisoning, and tool authorization.	2022–2026 Vienna, Austria
B.Sc. Physics Moscow Institute of Physics and Technology (MIPT)	Sept. 2018–June 2022 Moscow, Russia

Languages

Russian (native) | English (C1) | German (C1)